Physics 12 Winter 2026
Week 2 Discussion

TA: Jiashu Han
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1. **Summary of Topics from Last Week**

   - *From the lectures*
     - ◇ Scientific notation, SI units, unit conversion
     - ◇ Gaussian (normal) distribution
     - ◇ Exponential growth/decay, doubling time, half-life
     - ◇ Types of charts: linear, semi-log/log-linear, log-log
     - ◇ Order of magnitude (OOM) estimation
     - ◇ Newton's laws
   - *From the textbook* (Wolfson 4th edition)
     - ◇ Climate history of the Earth (Chapter 1)
     - ◇ Introduction to energy and power (first half of Chapter 2)
     - ◇ Energy usage in societies (second half of Chapter 2)

2. **Units, Scales, and Estimation**

   (a) **The "Human Power" Fermi Problem.**



   - Estimate how many joules of energy it takes to lift all students in this class from sea level to the summit of Mount Everest (approx. 8800 m). Express your final answer in scientific notation. *Hint*: Use $U = mgh$.

Solution: There are roughly 40 students in class. Assume on average, each student weighs around 70 kg, and we take $g \approx 10$ m/s^2. We can then find the gravitational potential energy gained by the group:

$$\begin{aligned} \Delta U &= (40 \times 70 \text{ kg}) \times 10 \text{ m/s}^2 \times 8800 \text{ m} \\ &\sim 3000 \text{ kg} \times 10 \text{ m/s}^2 \times 9000 \text{ m} \\ &\sim 3 \times 10^8 \text{ J} \end{aligned} \tag{1}$$

This is the amount of energy required to lift everybody up to the summit.

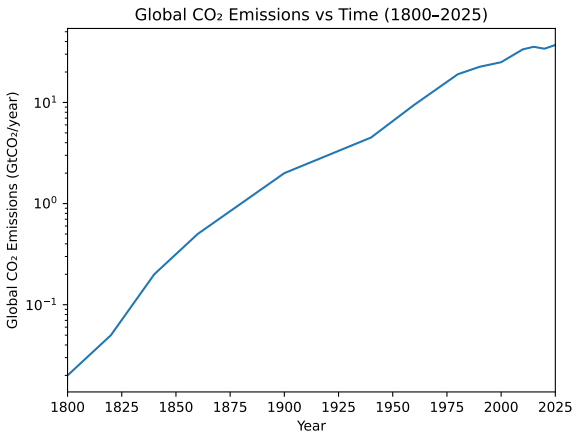
- Let's say that all of that lifting was done by 400 average energy workers, but they only work during the day. Roughly how many days would it take to lift everyone up?

  Solution: Recall from the textbook that an average energy worker outputs 100 J of energy per second. Since they only work during the day (assume that means 12 hours), the amount of energy all 400 workers output daily is

  $$W = 400 \times 100 \text{ J/s} \times 12 \text{ h} \times 3600 \text{ s/h} \sim 2 \times 10^9 \text{ J} \tag{2}$$

  Divide the amount of energy required by this, we get roughly 0.1 days, or a couple of hours. Depending how you choose to round the values, you may also get as high as half a day.

(b) **Reading Graphs**. Here's a plot showing global CO_2 emissions over time, from the beginning of the 19th century to the present.



- What did you notice on this plot?

  Solution: Many things: log-scale on the vertical axis but linear on the horizontal, roughly a straight line, years are discrete, etc.

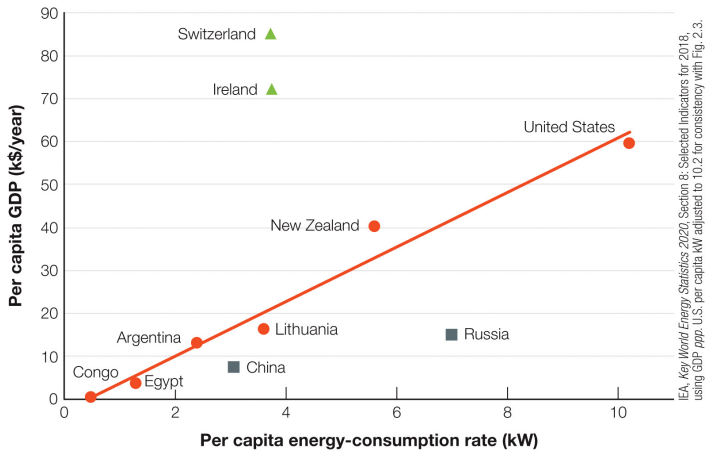
- If a line on a log-linear plot is straight, what does that tell you about the rate of growth? What about a straight line on a log-log plot?

  Solution:
  A straight line in a log-linear plot (like this one) represents exponential growth, since we are showing the logarithm of the actual function.

  On a log-log plot, a straight low represents a power law (i.e. linear, quadratic, cubic, etc.), depending on the slope of the line.

  To show this: the quantity on the vertical axis is $\log_{10} y$ and the quantity on the horizontal axis is the quantity on the vertical axis is $\log_{10} x$, so a straight line means $\log_{10} y = m \log_{10} x + b$, where m is the slope in the log-log plot. Apply an exponential of base 10 on both sides, we get $y = x^m + 10^b$, which is a power law, with the power determined by the slope in the log-log plot.

- Recall the following plot from the textbook. What does the slope of the red line roughly represent?

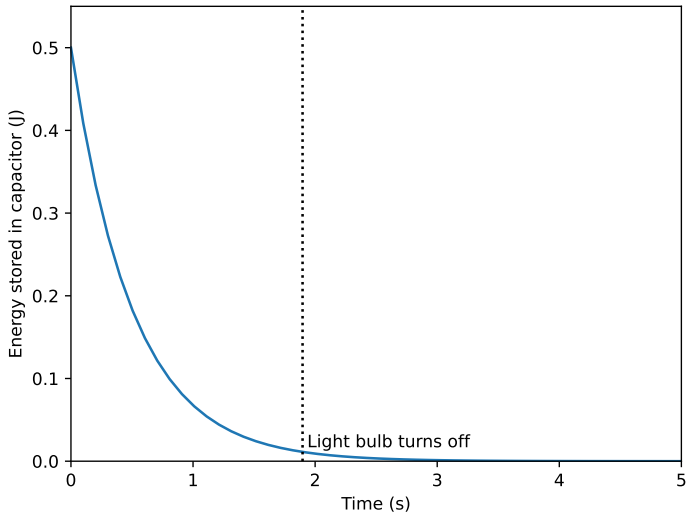


  Solution: It represents how much per capita GDP (measured in k$/year) is generated per kW of energy consumed. It's a measure of how efficient the economy is in energy usage. In contrast, energy intensity is

the inverse of the slope here, and quantifies the inefficiency of energy use.

3. **Mathematical Foundations**

(a) **The Slope of Energy**. A light bulb is connected to a fully charged capacitor (similar to a battery, but discharges quickly and cannot provide a steady flow of current) and a resistor. The light bulb gradually dims as the flow of charges diminishes. The following plot shows the energy E stored in the capacitor as a function of time t.



- What does the slope of this line represent physically? (Think about units).

  Solution: It represents the rate of energy flow out of the capacitor at that instant. If we assume all of that energy is flowing to the light bulb, the absolute value of the slope here would also be the power of the light bulb (not the rate power).

- If we instead connect the light bulb to a battery, which provides a steady flow of current to through the light bulb, what would this plot look like? What about its slope? What does that tell you about the power being delivered?

  Solution: It would look mostly flat, and the slope would be 0. This means the power being delivered to the light bulb is steady.

(b) **Compound interest**. You borrowed \$1000 from someone (probably black market) with a daily interest rate of 5%.

- If you don't pay off the debt, how many days would it take for the debt to double?
  Solution: Let's call the original debt D_0. Your debt grows like $D(t) = D_0 \cdot 1.05^{t/1\text{ day}}$, so we are looking for when $1.05^{t/1\text{ day}} = 2$, which is

$$t = \log_{1.05}(2)\text{ days} \approx 14.2\text{ days} \tag{3}$$

- Recall the formula for doubling time:

$$\text{doubling time} \approx \frac{70}{\text{rate in \%}} \tag{4}$$

  Check that this gives you the same result.

  Solution: Just plug in 5 at the bottom, and we get 14. This formula is accurate as long as the rate r is small compared to 1, as shown by the following steps (you are not required to know this)

$$\text{doubling time} = \log_{1+r}(2) = \frac{\ln 2}{\ln(1+r)} \approx \frac{\ln 2}{r} = \frac{100 \ln 2}{r\text{ in \%}} \tag{5}$$

  where $100 \ln 2 \approx 70$.

(c) **Bacteria Growth**. We are growing bacteria in a petri dish, and the condition is kept such that the bacteria will double their numbers after every 12 hours. At what daily rate is the number of bacteria increasing?

Solution: We are now given the doubling time T and want to find the growth rate. Since we want to find the *daily* growth rate, let's write the double time in terms of days: $T = 0.5$ days. The daily growth rate is therefore $2^{1\text{ day}/T} - 1 = 3 = 300\%$. Note that this rate is too big and therefore we cannot use the simple formula from the previous problem.

4. **Physics in Practice**



Terminal Velocity. The ball drop experiment may have been a myth, and could actually be a thought experiment. Regardless, let's drop a metal ball ourselves: We drop a ball of mass $m = 1.0$ kg from the 56 m tall tower, and the ball falls vertically downward through air. As the ball falls, two forces act on it: the downward gravitational force $F_g = mg$ and the upward air resistance (drag) $F_d = kv^2$. k is a positive constant that depends on the object's shape and the properties of air.

- Let's assume $k = 0.1$ kg/m. What is the ball's terminal velocity?

  Solution: When the ball reaches terminal velocity, the drag force has grown enough to be as big in magnitude as the gravitational force. Since the two forces point in opposite directions, the ball feels no net force and moves at constant speed. Therefore, we can set the two forces equal to each other: $mg = kv^2$. Plug in the values given above, and use $g = 9.8$ m/s^2, we get $v = \sqrt{mg/k} = 9.9$ m/s.

  Additional comments[1]: Since this is the maximum velocity the ball can attain, it means the ball will spend at least 56 m/(9.9 m/s) $\approx$ 6 seconds in the air, which is unrealistically long because of the value of k I chose (too high). This raises an important point: the ball drop experiment could actually show the two balls reach the ground at noticeably different times, if they have different terminal velocities and manage to reach a significant fraction of those velocities early in the fall. This means we can't blame Aristotle too much for not doing experiments — he may have chosen to

[1] Enrichment material.

drop a ball and a plate (or a piece of paper, basically anything that has a larger k and therefore different terminal velocity).

On the other hand, the rumored thought experiment consists of the following idea and is a much easier and more elegant way to prove Aristotle wrong: Consider two balls (one light and one heavy) chained and falling together. If Aristotle's claim was correct, the lighter ball would fall behind and drag the heavier back, making it fall more slowly. We could also consider the two balls together as a single object (since they are connected), but this object, being heavier, should fall faster. This contradiction leads to the conclusion that Aristotle's claim is false.

- *Challenge*[2]: How long would it take for the ball to reach terminal velocity? Would the ball reach terminal velocity at the bottom of the tower?

  Solution:

  According to Newton's second law, the acceleration of the ball (in the downward direction) is

$$a = \frac{\Sigma F}{m} = \frac{mg - kv^2}{m} = g - \frac{k}{m}v^2 = \frac{dv}{dt} \tag{6}$$

  Turns out we will have to solve a differential equation. Putting in the initial condition $v(0) = 0$ (i.e. initial velocity is 0), we find the solution

$$v(t) = v_t \tanh\left(\sqrt{\frac{gk}{m}}t\right) \tag{7}$$

  where $v_t = \sqrt{mg/k}$ is the terminal velocity we found earlier.

  If you look up what the tanh function looks like, you will find that $\tanh\left(\sqrt{\frac{gk}{m}}t\right)$ would approach 1 but never quite get there — The ball will never reach the terminal velocity, but only get close to it.

  We can, however, find how long it would take for the ball to reach 99% of its terminal velocity: $\tanh\left(\sqrt{\frac{gk}{m}}t\right) = 0.99$, which leads to $t = 2.67$ s. By this time, the ball has traveled about 19.6 m, so it will definitely reach 99% of the terminal velocity at the bottom.

[2]Note: I didn't actually think through this problem when I wrote it down and it's too challenging (sorry about that!). You do not need to know how to solve this part.